

Chapter One

Introduction

* Objective

- * To understand AI and related concept (Knowledge base and intelligent system)
- * Understand the components of AI system
- * Get a feel of application areas of AI
- * Get a feel of scholars view to define AI
- * Briefly discuss the difference between Expert system and other systems

Introduction

- * Intelligence is a general mental capability that involves the ability to reason, plan, solve problems, think abstractly, comprehend ideas and language, and learn.
- * Intelligence draws on a variety of mental processes, including memory, learning, perception, decision-making, thinking, and reasoning.

Human Intelligence

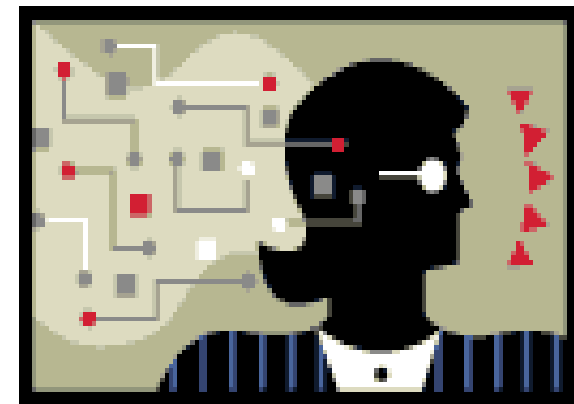
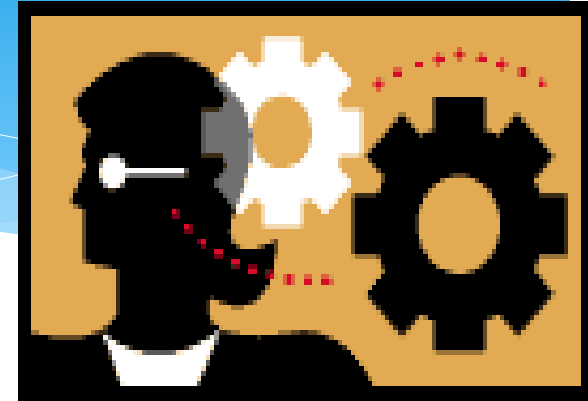
- ❑ How does our brain work when we solve a problem?
 - ❑ Do we think it over and suddenly find an answer?
 - ❑ What do we do when solving a complicated factorization problem, a puzzle or a mystery?

How do people Reason?

- They create categories and relate one concept to another: **Reasoning by analogy**:
 - * From the fact that: A is like B. M is in A. N is in B. Can we say “M is like N”?
- They use specific rules : **rule-based reasoning**
 - if ‘a’ then ‘b’ and if ‘b’ then ‘c’ $\rightarrow$ THEN $a \rightarrow b \rightarrow c$
- They use Past Experience – “CASES” (**case-based reasoning**)
 - Similarities of current and previous case
 - Store cases using key attributes
- They Use “Expectations” (**probabilistic reasoning**)
 - With the help of previous experience they use Heuristics - “Rule of thumb” to guess the probability of occurrence of events or guess solution for the problem at hand.

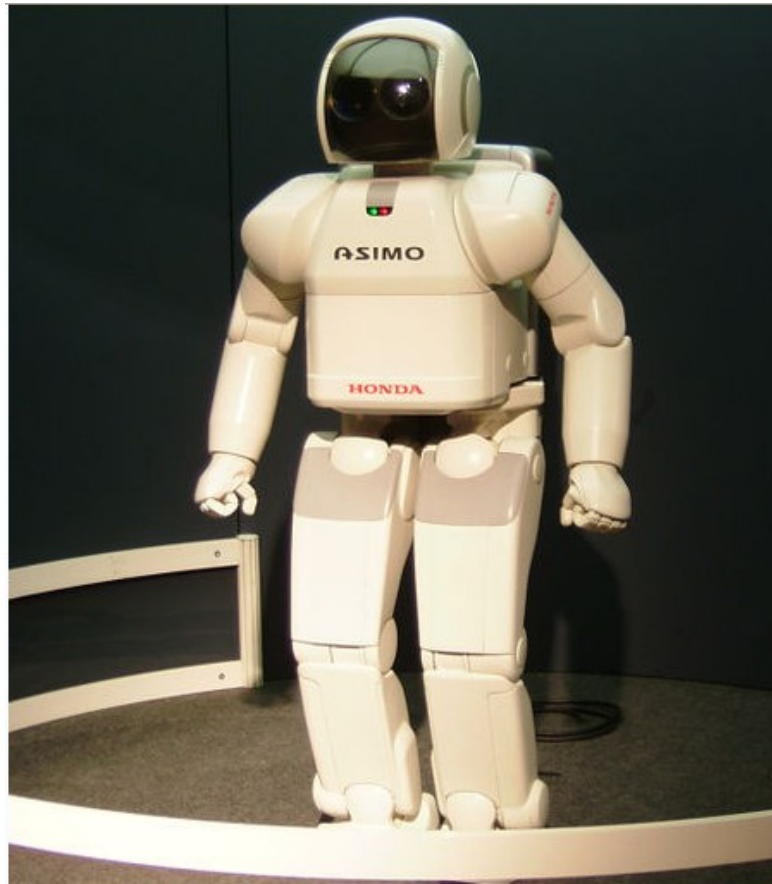
Deductive and inductive reasoning

- * **Deductive reasoning:** exact reasoning. It deals with **exact facts and exact conclusions**
 - * Gravity makes things fall. Hence, the apple that hit my head was due to gravity.
- * **Inductive reasoning:** reasoning from a set of facts or individual cases to a **general conclusion**
 - * How much to Buy ? The base costs is X, extra cost is Y, plus tax is at Z. Hence, it is better to deal at XYZ.
 - * Can we cut cost ? Heating cost X, lighting Y, parts Z, total cost is XYZ which is greater than our revenue R. Hence, we must cut costs.



Introduction

The term AI is first used by John McCarthy (1956) who considers it to mean the science and engineering of making intelligent machine.



Introduction

- * AI system is used in control, planning, and scheduling, recognition of handwriting, natural language, speech and face
- * Memory of Intelligent system is used to store knowledge base which is the key for success for artificial intelligent systems.
- * AI attempt to understand intelligent entities as well as to build them in order to understand human behavior.
- * The idea is to develop an intelligent systems that has human level intelligence or better if possible.

Introduction cont...

* AI system possess

- * **Natural language processing** to enable it to communicate successfully in human language
- * **Knowledge representation** to store what it knows or hears
- * **Automated reasoning** to use the stored information to answer questions and to draw new conclusions
- * **Machine learning** to adapt to new circumstances and to detect and extrapolate patterns
- * **Computer vision** to perceive objects
- * **Robotics** to manipulate objects and move about
- * And others

Views of defining AI

- What is AI (Artificial Intelligence)
 - * Different scholars define AI differently

(A) AI as a system
that think humanly

Concerned with
thought processing
and reasoning

(C) AI as a system
that Act humanly

(B) AI as a system
that think rationally

Concerned with
behaviors of agents

(D) AI as a system
that Act rationally

Views of defining AI

measures success of AI in
terms of human being
performance

(A) AI as a system
that think humanly

(C) AI as a system
that Act humanly

measures success of AI in
terms of ideal concept of
intelligence (rationality)

(B) AI as a system
that think rationally

(D) AI as a system
that Act rationally

Views of defining AI

- AI as systems that think humanly

- * “The automation of activities that we associate with human thinking, activities such as decision-making, problem solving, learning ...” (**Bellman** definition, 1978)
- * “The exciting new effort to make computers think ... machines with minds, in the full and literal sense” (**Haugeland** definition, 1985)

Thinking humanly: cognitive modeling

* Requires:

- * Scientific theories of internal activities of the brain

- * How human thinks?

- * How to validate a given agent think humanly?

- * The answer Requires either

1. Predicting and testing the behavior of human subjects from his/her thinking point of view (top-down) or
2. Direct identification from neurological data (bottom-up)

- * Study on Mental processing logic of human being (cognitive science) is not yet fertile

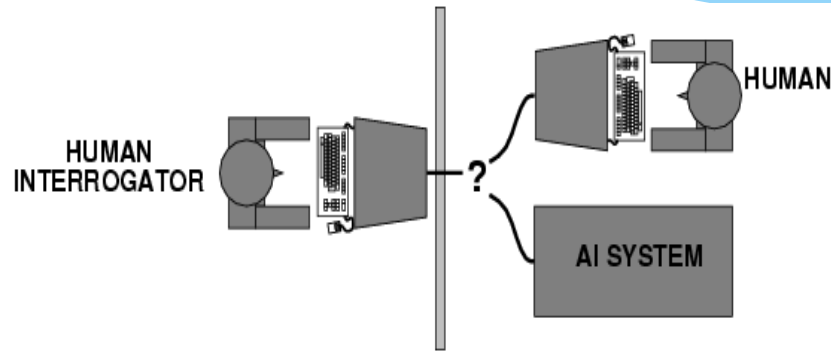
Views of defining AI

- * AI as systems that act humanly

- * “The art of creating machines that perform functions that require intelligence when performed by people.” (Kurzweil definition, 1990)
- * “The study of how to make computers do things at which, at the moment, people are better.” (Rich and Knight definition, 1991)

Acting humanly: Turing Test

- ❖ Turing (1950) on his famous paper "Computing machinery and intelligence":
- ❖ "Can machines think?" → "Can machines behave intelligently?"
- ❖ Operational test for intelligent behavior: the Imitation Game



- ❖ Predicted that by 2000, a machine might have a 30% chance of fooling a person for 5 minutes
- ❖ Anticipated all major arguments against AI in following 50 years
- ❖ Active areas of research to achieve this: Machine learning, NLP, Computer vision, etc

Introduction cont...

- * AI as systems that think rationally
 - A system is said to be rational if it does the “right thing” given what it knows.
- * “The study of mental faculties through the use of computational models.” (Charniak and McDermott definition, 1985)
- * “The study of the computations that make it possible to perceive, reason, and act.” (Winston definition, 1992)

Thinking rationally: "laws of thought"

- * Right thinking is related to irrefutable reasoning process
- * Require structure that always gave correct conclusion given correct premises
- * Logic is the key to design and implement an agent that think rationally
- * Several Greek schools developed various forms of *logic: notation* and *rules of derivation* for thoughts;
- * Direct line through mathematics and philosophy to modern AI
- * **Problems:**
 1. Not all intelligent behavior is mediated by logical deliberation (unable to take informal knowledge for decision making process)
 2. What is the purpose of thinking? What thoughts should I have? (there is a big difference between being able to solve a problem “in principle” and doing so “¹⁵in practice”)

Introduction cont...

- * AI as systems that act rationally
 - * “Computational intelligence is the study of the design of intelligent agent” (Poole, et al definition, 1998)
- * “AI ... is concerned with intelligent behavior in artifacts.” (Nilsson definition, 1998)
- * The course advocates to agents that act rationally

Acting rationally: rational agent

- * Means acting so as to achieve one's goals, given one's beliefs.
- * In this approach, AI is viewed as the study and construction of rational agent
- * Rational behavior: doing the right thing
- * The right thing: is the action/decision which is expected to maximize goal achievement, given the available information
- * Doesn't necessarily involve thinking
 - * e.g., blinking reflex
- * One way of acting rationally is to reason logically to the action. This indicates, making correct inference is part of being a rational agent
- * But rationality doesn't require correct inference because some time without having correct thing to do, agent must act rationally

Introduction cont...

- * AI follows all the four approach but tension existing between approaches which are centered around human and rationality.
- * Human centered approach must be empirical science, involving **hypothesis** and **experimental confirmation**.
- * A rationalist approach involves a combination of **mathematics** and **engineering**
- * This course focus on the study of a rational agent that think and act rationally.

Some Application areas of AI

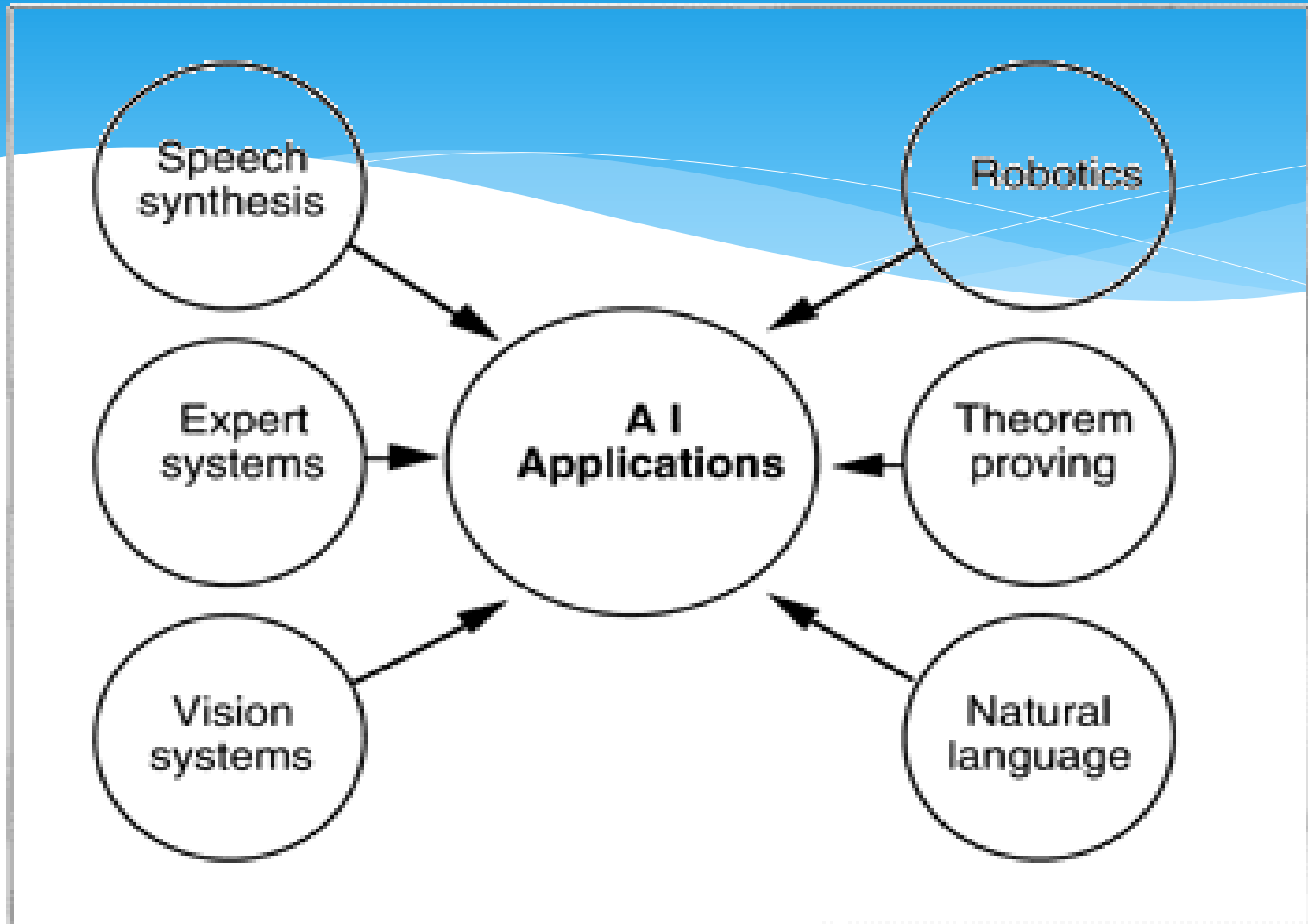


Figure 1. Some applications of Artificial Intelligence

AI prehistory

- * Philosophy Logic, methods of reasoning, mind as physical system foundations of learning, language, rationality
- * Mathematics Formal representation and proof algorithms, computation, (un) decidability, (in) tractability, probability
- * Economics Decision theory
- * Neuroscience physical substrate for mental activity
- * Psychology phenomena of perception and motor control, experimental techniques
- * Computer engineering building fast computers
- * Control theory design systems that maximize an objective function over time
- * Linguistics knowledge representation, syntax, grammar

History of AI

* **Warren McCulloch and Walter Pitts (1943)**

- * 1st AI work: Boolean circuit model of the brain
- * Drew on three sources
 1. Knowledge of the basic physiology and function of neurons in the brain
 2. The formal analysis of propositional logic due to Russell and Whitehead
 3. Turing's theory of computation
- They proposed a model of artificial neurons
- They showed any computable function could be computed by some network of connected neurons
- They also suggested that suitably defined networks could learn

History of AI

- * **Claude Shannon (1950) and Alan Turing (1953)**

- Write a chess program

- **Marvin Minsky and Dean Edmonds (1951)**

- Built the 1st neural network computer

History of AI Cont..

- * Newell and Simon develop a reasoning program called the **Logic Theorist (LT)** before Dartmouth workshop
- * They then come up with the **General Problem Solver (GPS)**
- * **GPS**, unlike **LT**, is designed to imitate human problem solving protocols and it is the 1st program to embody the “thinking humanly” approach of AI
- * Herbert Gelernter, 1959 constructed the **Geometry Theorem Prover (GTP)**

History of AI Cont..

- * Arthur Samuel, 1952 wrote a series of checker programs
 - * It can learn, which disprove the idea that computer can only do what they are told to do
 - * His program play better than the creator
- * John Mcarty, 1958 at MIT
 - * Define the 2nd old high level programming language, LISP which is the 1st and dominant AI programming language
 - * Invent time sharing concept with his friend to avoid the problem of time as a resource.

History of AI Cont..

- * 1958 is also marked the year that Marvin Minsky moved to MIT
- * Minsky supervised a series of students who choose limited problems that appeared to require intelligence to solve (this problems known as **Micro world**)
- * The most dominant micro world problem is the block world
- * It consists of a set of solid blocks placed on top of the table

Summary on history of AI

- * 1943 McCulloch & Pitts: Boolean circuit model of brain
- * 1950 Turing's "Computing Machinery and Intelligence"
- * 1956 Dartmouth meeting: "Artificial Intelligence" adopted
- * 1952—69 Look, Ma, no hands!
- * 1950s Early AI programs, including Samuel's checkers program, Newell & Simon's Logic Theorist, Gelernter's Geometry Engine
- * 1965 Robinson's complete algorithm for logical reasoning
- * 1966—73 AI discovers computational complexity
Neural network research almost disappears
- * 1969—79 Early development of knowledge-based systems
- * 1980-- AI becomes an industry
- * 1986-- Neural networks return to popularity
- * 1987-- AI becomes a science
- * 1995-- The emergence of intelligent agents

Basic Kinds of Systems

- * System is a set of components that interact to each other in a logical way to achieve specific goals.
- * There are different types of system based on the services, the user type, and the method of operations
- * Some of the systems includes:
 - * Information Systems
 - * Database Management System
 - * Information Retrieval System
 - * Expert System

* Information Systems

- * Is a system whether automated or manual, that comprises people, machines, and/or methods organized to collect, process, transmit, and disseminate data that represent user information
- * Provides the information to the appropriate user at the right time and place.

* Data Base Management System

- * DBMS is a software designed for the purpose of managing a database
- * A database is a structured collection of records or data that is stored in a computer so that a program can consult it to answer queries.

* Information Retrieval System

* It is the science of

- searching for **information** in documents,

- * searching for documents themselves

- * searching for metadata which describe documents, or

- * searching within database, whether relational stand-alone databases or hyper textually-networked databases such as the World Wide Web.

* It is used to reduce information overloading

Expert system

- * An **expert system**, also known as a knowledge based system, is a computer program that contains some of the subject-specific (domain specific) knowledge of one or more human experts.
- * Is a system with two basic components:
 - * Knowledge base, which model the knowledge of an expert in the area under consideration
 - * Inference engine
- * When it is used by non expert user, it can serve as an expert that guide the user to make an expert decision. (doctors, engineers, lawyers, etc)

Expert system

- * **Examples:**
 - * Dendral, MYCIN, PUFF, ELIZA, BTDS, etc.
-
- * Dendral expert system:
 - * The primary aim to aid organic chemists with identification of unknown organic molecules by analyzing information from mass spectrometry graphs and the knowledge of chemistry

Expert system

* MYCIN:

- * Written in LISP around 1970s and derived from Dendral expert system
- * was designed to diagnose infectious blood diseases and recommend antibiotics, with the dosage adjusted for patient's body weight
- * It would query the physician/patient running the program via a long series of simple yes/no or textual questions.
- * At the end, it provides
 - * a list of possible cause bacteria ranked from high to low based on the probability of each diagnosis,
 - * its confidence in each diagnosis' probability
 - * the reasoning behind each diagnosis
- * It has around ~50 rules

Expert system

* PUFF:

- * PUFF can diagnose the presence and severity of lung disease and produce reports for the patient's file
- * Is an Expert System that interprets lung function test data and has become a working tool in the pulmonary physiology lab of a large number of hospital
- * Puff went into production at Pacific Medical Center in San Francisco in 1977.
- * Several hundred copies have been sold and are in use around the world.

Expert system

* ELIZA :

- * ELIZA is a very well-known artificial intelligence program designed to emulate a Rogerian psychotherapist.
- * The basic elements of Carl Rogers' new way of therapy was to have a more personal relationship with the patient, to help the patient reach a state of realization that they can help themselves
- * ELIZA was showcased for a number of years at the **MIT AI Laboratory**.
- * ELIZA has no reasoning ability, cannot learn
- * ELIZA only appears to understand because "she" uses canned responses based on keywords, as well as string substitution
- * Visit and interact with ELIZA at <http://www-ai.ijs.si/eliza/eliza.html>

Expert system

* ELIZA :

* **Eliza:** Hello. I am ELIZA. How can I help you?

* **You:** I am Samuel

* **Eliza:** How long have you been Leykun?

* **You:** ten years

* **Eliza:** Please go on.

* **You:** do you know Ethiopia

* **Eliza:** Oh, i know ethiopia.

* **You:** do you understand Amharic

* **Eliza:** Oh, i understand amharic.

Strong AI vs. Weak AI

AI is the concept that it is possible for a machine to think in the same sense as human do.

- * **Strong AI:** argues that is possible that one day a computer will be invented which can be called a **mind** in the fullest sense of the word. That means,
 - * Computers can think, reason, imagine, etc. and do all the things that we currently associate with human brain.
- * **Weak AI:** argues that computers can only appear to think and are not actually conscious in the same way as human brains are.



* AI Demo